ALBENDAZOLE

Avoid in pregnancy
Fatty meal for tissue infections

CHEMICAL NAME Methyl N-(6-propylsulfanyl-1H-benzimidazol-2-yl)carbamate
CLASS Benzimidazole anthelmintic (broad spectrum)
TRADE NAMES Zentel · Albenza · Alzental

THERAPEUTIC USES / INDICATIONS

- Intestinal worms: ascariasis, hookworm, threadworm (enterobiasis), trichuriasis, strongyloidiasis.
- Hydatid disease (echinococcosis); with or before surgery.
- Neurocysticercosis.
- Mass deworming programmes (school children).

ADMINISTRATION

- **PO**; tablets can be chewed or crushed.
- Intestinal worms: take on empty stomach or with food.
- Tissue infections (hydatid, neurocysticercosis): with a fatty meal (↑ absorption 5×).

DOSAGES

- **Intestinal worms (adult & child >2 yr):** 400 mg single dose (children 1–2 yr: 200 mg).
- **Threadworm:** 400 mg, repeat after 2 weeks.
- **Hydatid disease:** 400 mg twice daily (15 mg/kg/day if <60 kg) for 28-day cycles with 14-day breaks.
- **Neurocysticercosis:** 400 mg twice daily for 8–30 days (with steroids).

PHARMACOKINETICS

- Poorly absorbed (helpful for gut worms); absorption ↑ with fat.
- Rapid first-pass to active albendazole sulfoxide.
- Excreted in bile and urine. **t½:** 8–12 h (sulfoxide).

PHARMACODYNAMICS

- Binds parasite **β-tubulin** → inhibits microtubule polymerisation → impaired glucose uptake → energy depletion → death of worm.
- Vermicidal, ovicidal, larvicidal.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Abdominal pain, nausea, diarrhoea.
- Headache, dizziness.
- Reversible hair loss (long courses).

ADVERSE EFFECTS (SERIOUS)

- Hepatotoxicity (raised enzymes) with long courses.
- Bone-marrow suppression (leucopenia, agranulocytosis); long courses.
- Neurocysticercosis: seizures, raised ICP from dying cysts (give steroids).

CONTRAINDICATIONS & CAUTIONS

- Pregnancy (teratogenic in animals) — especially 1st trimester; exclude pregnancy before long courses.
- Hypersensitivity to benzimidazoles.
- **Caution:** liver disease, bone-marrow depression.

KEY INTERACTIONS

- **Dexamethasone, praziquantel, cimetidine:** ↑ albendazole sulfoxide levels.
- **Carbamazepine, phenytoin:** ↓ levels.

NURSING MANAGEMENT

- Check pregnancy status; advise contraception during and 1 month after long courses.
- Long courses: CBC and LFTs at start and every 2 weeks.
- Neurocysticercosis: monitor neurological status; steroids/antiepileptics as ordered.
- Threadworm: treat whole household.
- Hygiene education.

PATIENT EDUCATION

- Wash hands after the toilet and before eating; keep nails short.
- Threadworm: treat the whole family; wash bedding and towels.
- For long courses, take with fatty food and attend blood tests.
- Avoid pregnancy during treatment.

KEY POINTS

- Benzimidazole — inhibits worm β-tubulin.
- Fatty meal for hydatid/neurocysticercosis.
- 400 mg single dose for gut worms.
- Avoid in pregnancy; CBC/LFTs in long courses.